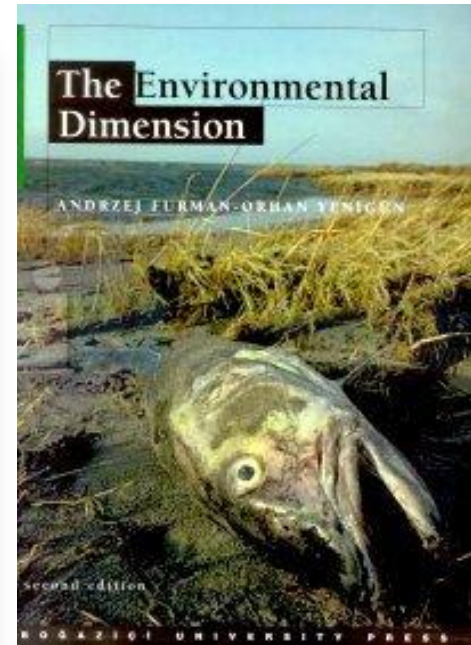


ESc 301.03

The Environmental Dimension

Assoc. Prof. Dr. Başak Güven



Reference textbook

Instructor:

Assoc. Prof. Dr. Başak Güven

Boğaziçi University, Institute of Environmental Sciences

e-mail: basak.guven@bogazici.edu.tr

Office Hours: Students can contact the lecturer by e-mail for suitable hours.

Course Description:

The purpose of this course is to introduce current environmental topics and elements of ecology to a wide spectrum of students with technical as well as social background.

Course Structure:

3 hours of lecture/week. TTTh (455)
HD 201

Exams and Quizzes:

- ❖ 2 Midterm Exams (25 % each)
- ❖ 1 Final Exam (35 %)
- ❖ Quizzes (15 %) (During the course approximately 5 unannounced quizzes will be handed out).

Optional: Students are encouraged to give a 10-15 min. presentation on a topic they choose. This will bring up to 4 extra points.

Attendance is strongly advised for this course. Students are advised to attend at least 60 % of the lectures. Attending the course very regularly (80 % and over) will increase the chance of receiving a better grade.

Essential are my Power Point slides that I present in class. I also follow some chapters of the textbook 'The Environmental Dimension' which is written by our Institute members.

Course Outline

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graph TD; A[Course Outline] --> B[Part I: Ecology]; A --> C[Part II: Environmental Problems];
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Part I: Ecology

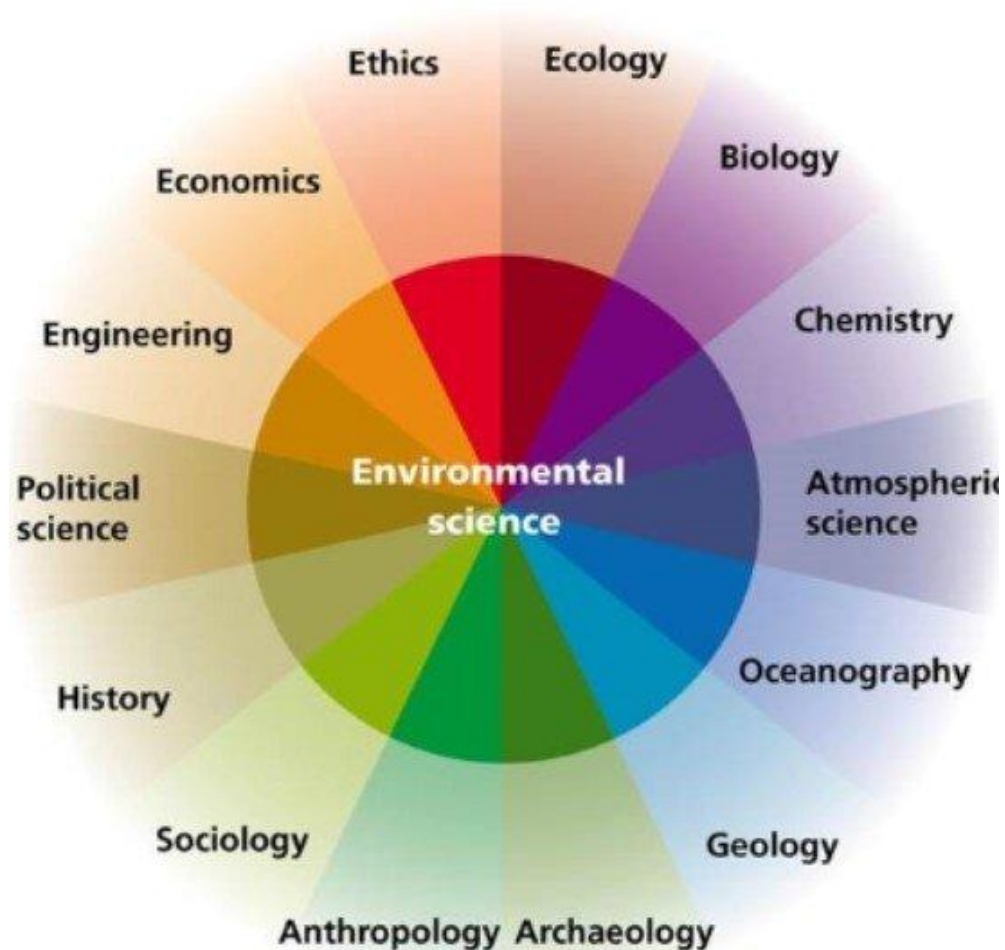
1. **Introduction to Ecology** - species, natural selection, adaptation, extinction, evolution of life
2. **Populations and Communities** - population growth, biotic potential, environmental resistance, survivorship, age structure, ecological niche, edge effect, interspecific relationships, ecological succession
3. **Ecosystems** - ecospheres, producers, consumers, decomposers and detritivores, trophic level, food chain
4. **Energy in Ecosystems** - energy flow through ecosystems, energy budget, the law of 10 %, pyramids of energy and biomass, primary productivity
5. **Biodiversity** - genetic, species and ecosystem diversity, benefits of biodiversity, threats to biodiversity
6. **Biomes** - terrestrial and aquatic biomes
7. **Material Cycles** - water cycle, carbon cycle, nitrogen cycle, phosphorus cycle, sulfur cycle, human interference in cycles, The Gaia Theory

Part II: Environmental Problems

1. **Population Growth, Agriculture and Pesticides** - human population growth, industrial revolution, population projections, population growth and age structures in Turkey, population growth and environmental problems, agriculture, food supply, green revolution, resource use in developed and developing countries; soil pollution; pest and weed control, chemical pesticides, health effects, biomagnification, biological pest control, integrated pest management
2. **Energy Problems** - energy supplies, energy from fossil fuels, fossil fuel fired power plants, nuclear energy, nuclear power plants, nuclear energy problems, exposure to radiation, renewable energy sources, direct solar energy, indirect forms of solar energy (wind, water, biomass), geothermal energy, tidal energy, ocean thermal energy
3. **Water Resources, Water Pollution and Control** - water resources, water consumption, sources of pollution, water quality problems, water pollutants, water quality parameters (DO, BOD, COD), surface water pollution, groundwater pollution, drinking water treatment, wastewater treatment, water reclamation and reuse, coastal pollution, oil spills, water pollution control regulations
4. **Air Pollution and Control** - atmospheric structure, sources of air pollution, primary and secondary air pollutants, health problems, acid rain, thermal inversion, winter smog, photochemical smog, indoor air pollution, air pollution control technologies, air pollution control regulations
5. **Noise Pollution** - definition and unit of noise, sources and effects
6. **Global Atmospheric Pollution** - solar energy balance, greenhouse effect, greenhouse gases, enhanced greenhouse effect, sources and emissions of greenhouse gases, global warming, positive and negative feedbacks, solutions to global warming; ozone depletion, formation and depletion of the ozone layer, ozone depleting chemicals, ozone hole problem, consequences of ozone depletion, solutions to ozone depletion
7. **Solid Waste Generation and Disposal** - (generation and composition of solid waste, waste management hierarchy, municipal waste landfilling, incineration, recycling, hazardous wastes, hazardous waste landfilling, medical wastes, characteristics of plastics, life cycle assessment, comparison of plastics and paper, solid waste regulations
8. **Sustainable Development** - earth's carrying capacity, overview of environmental issues, sustainable development, main goals, ecological industrial network

WEEKLY OUTLINE

Date	Subject
10.02.2026 (T)	Introduction to Environmental Sciences: What is Environmental Science?
12.02.2026 (Th)	Introduction to Ecology
17.02.2026 (T)	Populations & Communities
19.02.2026 (Th)	Ecosystems
24.02.2026 (T)	Biodiversity
26.02.2026 (Th)	Materials Cycles I
03.03.2026 (T)	Materials Cycles II
05.03.2026 (Th)	Population Growth
10.03.2026 (T)	Agriculture, Soil Degradation, Pollution
12.03.2026 (Th)	Pesticides
17.03.2026 (T)	1 st Midterm
24.03.2026 (T)	Energy Problems – Fossil Fuels
26.03.2026 (Th)	Energy Problems – Nuclear Energy
31.03.2026 (T)	Energy Problems – Renewable Energy
02.04.2026 (Th)	Water Quality I
07.04.2026 (T)	Water Quality II
09.04.2026 (Th)	Water Treatment
14.04.2026 (T)	Wastewater Treatment
16.04.2026 (Th)	Air Pollution & Noise
28.04.2026 (T)	Global Atmospheric Pollution
30.04.2026(Th)	2 nd Midterm
05.05.2026 (T)	Solid Waste
07.05.2026 (Th)	Sustainable Development
12.05.2026 (T)	Presentations
14.05.2026 (Th)	TBA



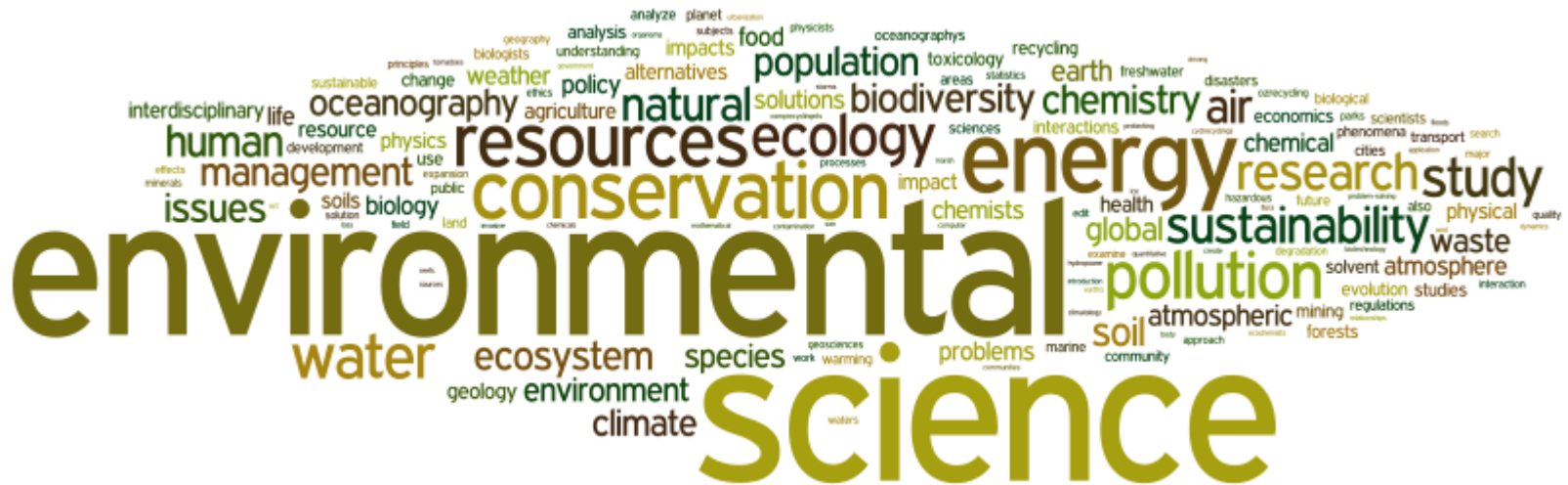
CHEMICAL
ENGINEERING
CHEMISTRY
CIVIL ENGINEERING
COMPUTER
ENGINEERING
ECONOMICS
ELECTRICAL &
ELECTRONICS
ENGINEERING
FOREIGN LANGUAGE
EDUCATION
GUIDANCE &
PSYCHOLOGICAL
COUNSELING
HISTORY
INDUSTRIAL
ENGINEERING
INTERNATIONAL
TRADE
LINGUISTICS
MANAGEMENT
MOLECULAR
BIOLOGY &
GENETICS
PHILOSOPHY
PHYSICS
PSYCHOLOGY
SOCIOLOGY
TEACHING
CHEMISTRY PR.
TRANSLATION AND
INTERPRETING
STUDIES
TURKISH LANGUAGE
& LITERATURE
UNDERGRADUATE
PROGRAM IN
SCIENCE EDUCATION
WESTERN
LANGUAGES &
LITERATURES

ESc 301.03

The Environmental Dimension

Introduction to Environmental Sciences

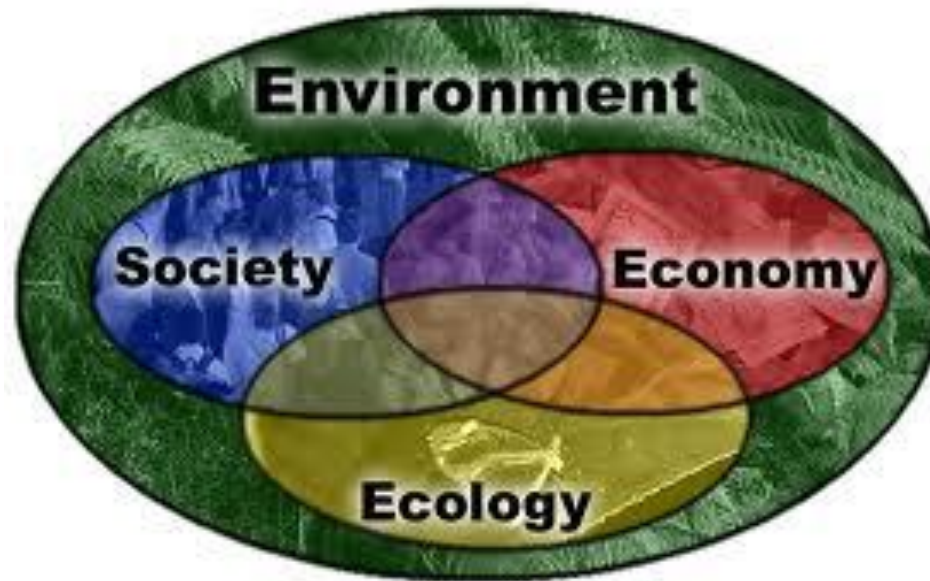
What is Environment?



Dr. Başak Güven

Environmental Science/Studies

Interdisciplinary studies in natural sciences, including geology, climatology, hydrology, ecology, **and their interaction** with social sciences such as economics, political science, sociology, anthropology, geography.

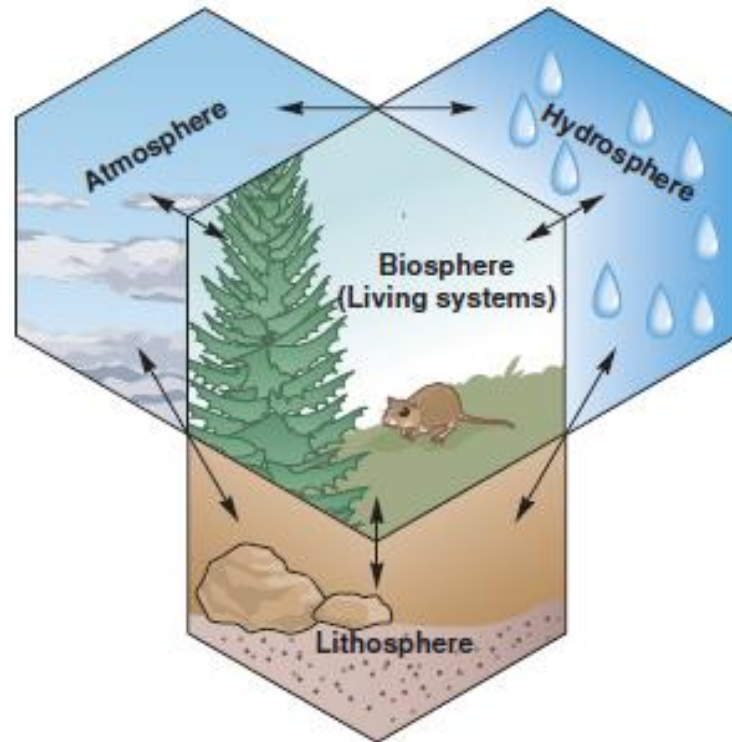


http://commons.wikimedia.org/wiki/File:Ecology_Society_Economy_diagram_Environment_background.jpg

What is Environment?

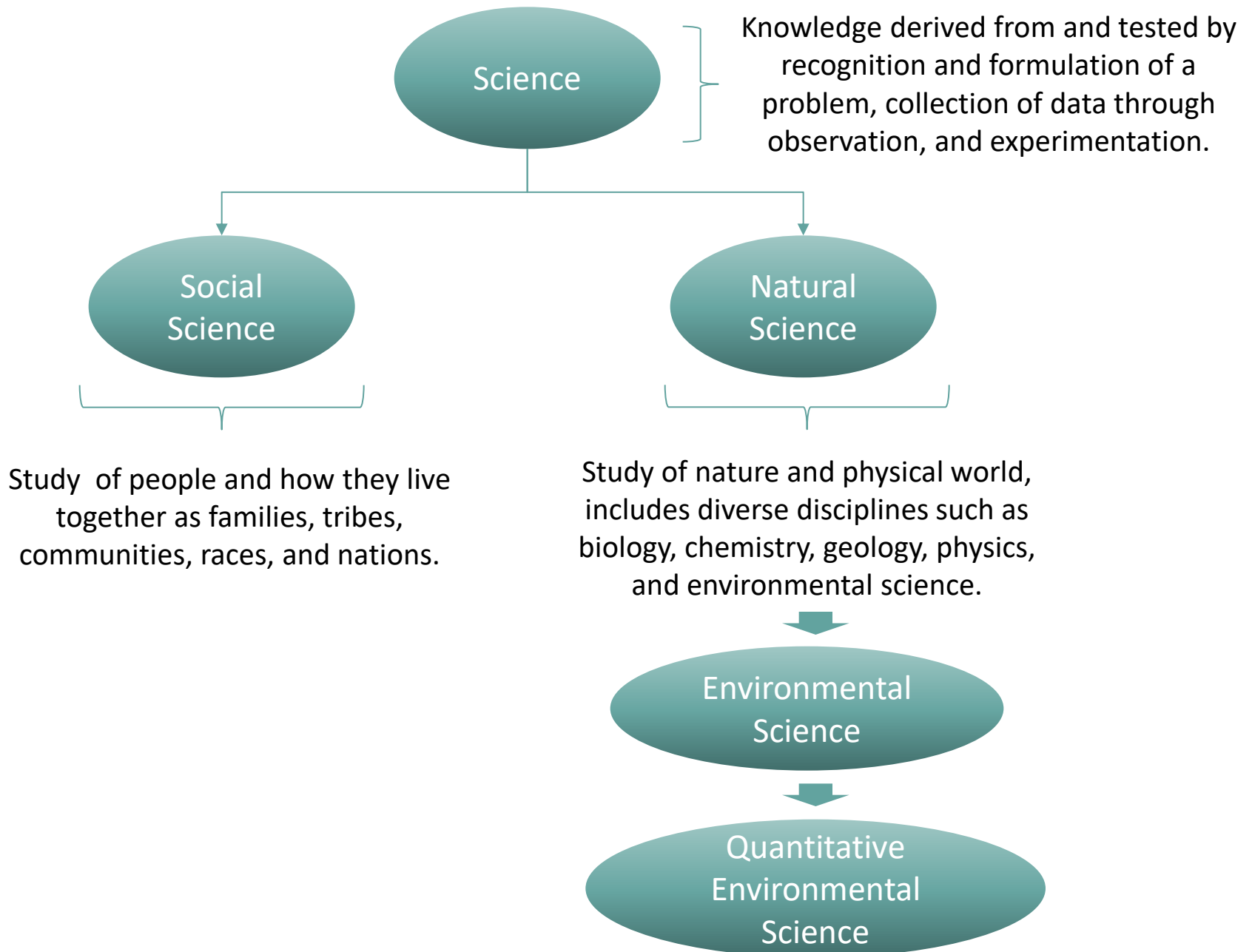
The environment encompasses the whole of life on earth and the complex interactions that link the living world with the physical world.

In a general sense, this covers everything contained within the air, land and water.



A collection of all the Earth's ecosystems:

- ❖ Biosphere
- ❖ Hydrosphere
- ❖ Atmosphere
- ❖ Lithosphere



The Nature of Research

e.g.
Disciplines

Biology

Chemistry

Physics

e.g.
Subdisciplines

Microbiology

Organic chemistry

Nuclear physics

Environmental sciences
encompasses all the
fields of natural science

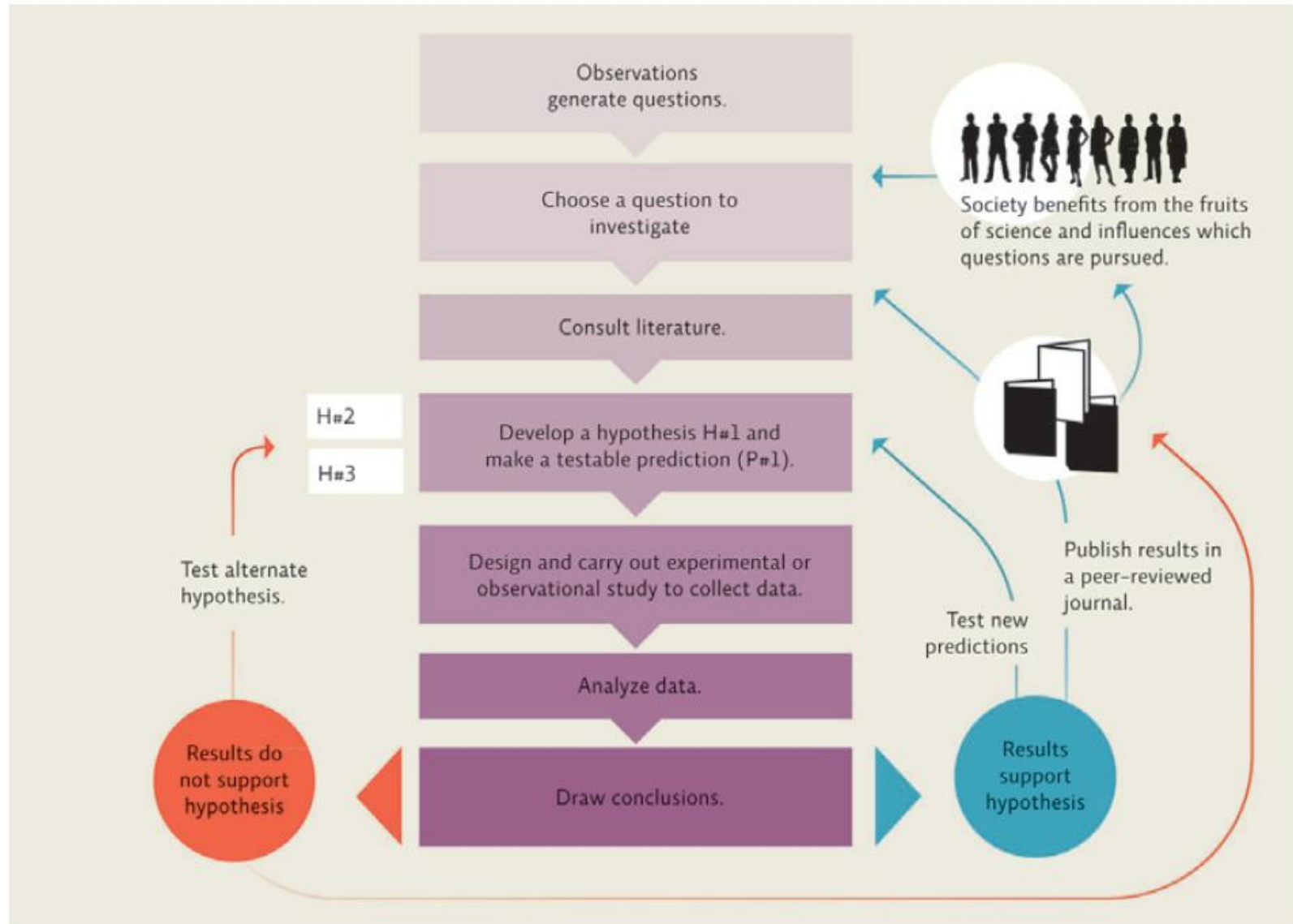
- the atmosphere
- the land
- the water
- and their inhabitants (as differentiated from the built environment)

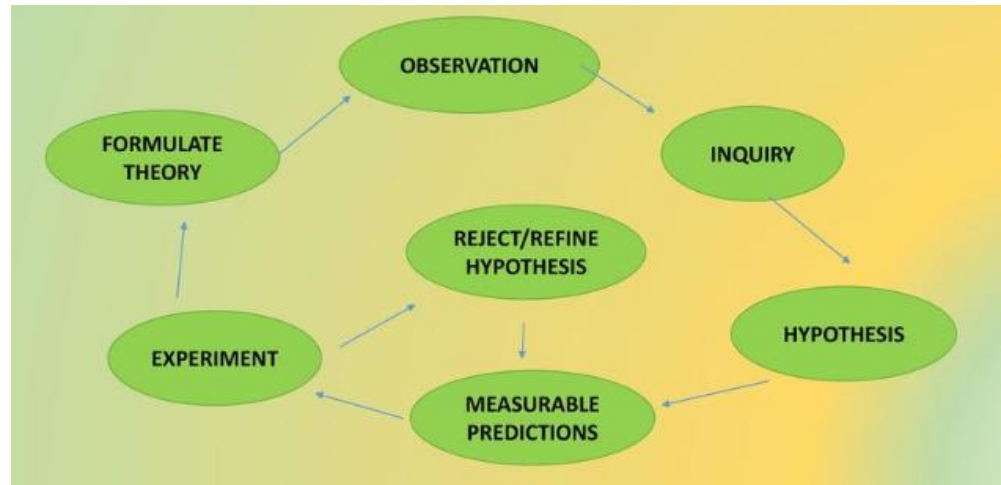
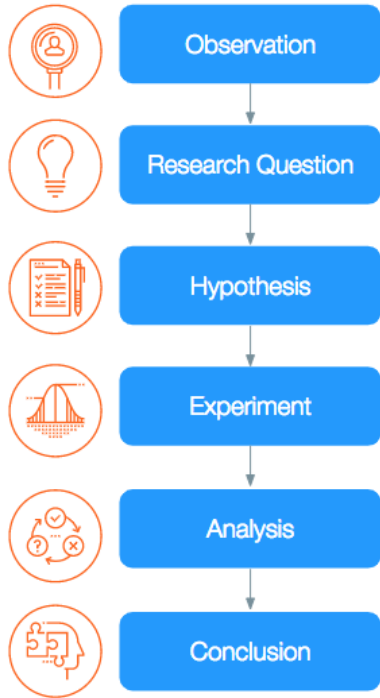
The global environment consists of the atmosphere, the hydrosphere, and the lithosphere in which the life-sustaining resources of the earth are contained.

Modern environmental science has also found applications to the built environment or, perhaps more correctly, to the effusions from the built environment.

The Scientific Method

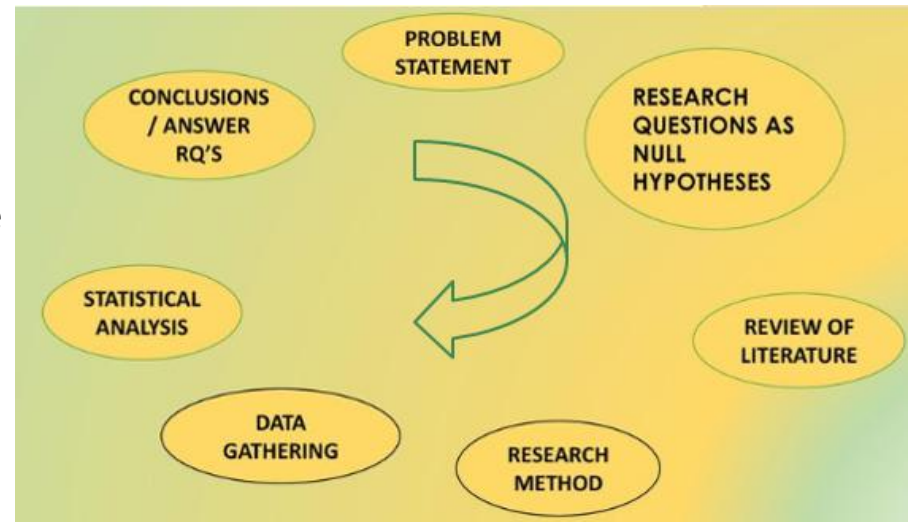
↓ Scientists work from previous knowledge and observation to ask new questions and pose possible explanations (hypotheses) for what they observe. They then design a study to gather evidence to test predictions made from their hypotheses. The scientific method is not necessarily a linear sequence. Rather, it is more of a cycle that scientists move through in whatever order of “steps” best suits their needs.





Environmental sciences research is traditionally carried out through the accumulation of observations of systems and system behaviour under ‘**natural**’ circumstances and during ‘**experimental**’ manipulation..

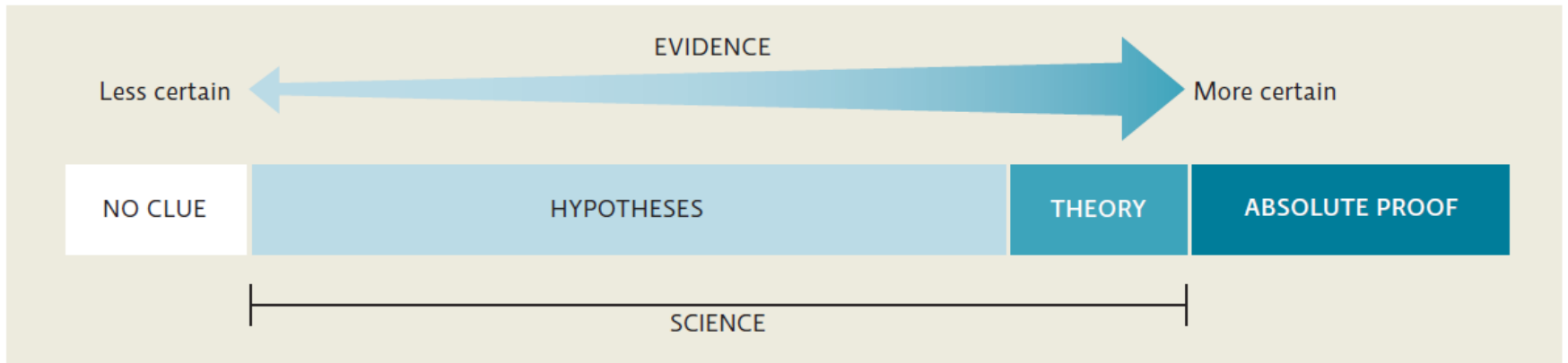
These observations provide the evidence upon which hypotheses can be generated about the structure and function of the systems. These hypotheses can be tested against new observations and, where they prove to be reliable descriptors of the system behaviour, then they may eventually gain recognition as tested theory or general law.

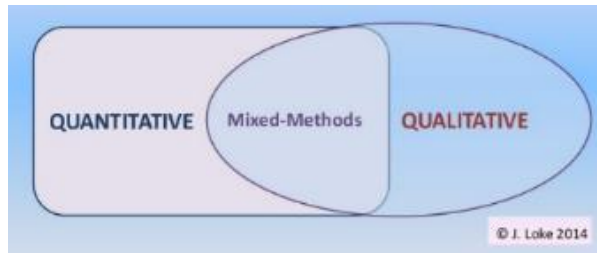


The conditions, which are required to facilitate research, include:

- ❖ a means of observation and comparative observation (measurement)
- ❖ a means of controlling or forcing aspects of the system (experimentation)
- ❖ an understanding of previous research and the state of knowledge (context)
- ❖ a means of connecting threads of above matters (imagination)

↓ There are degrees of certainty in science; we know some ideas are better than others. The more evidence we have in support of an idea, especially when the evidence comes from different lines of inquiry, the more certain we are that we are on the right track. But since all scientific information is open to further evaluation, we do not expect or require “absolute” proof.





PARAMETERS	QUANTITATIVE RESEARCH	QUALITATIVE RESEARCH
GENERAL NATURE	Objective approach to seek precise measurement in numerical form	Subjective approach to seek in-depth description in narrative form
KNOWLEDGE OF STUDY VARIABLE	Variables are clearly understood & defined in advance by the researcher	Researcher may have only rough idea about variables in advance.

Discrete Continuous



A sample of the universe of possibilities

Hypothesis

one or more hypotheses that have been supported by repeated testing

Theory

Law

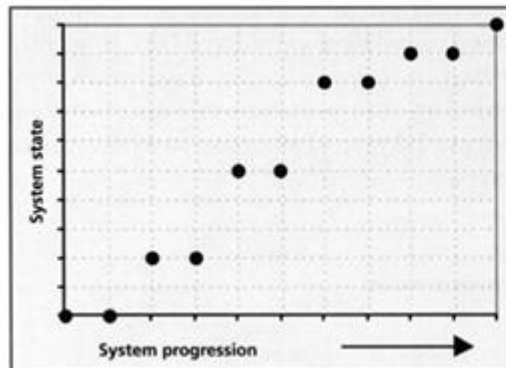
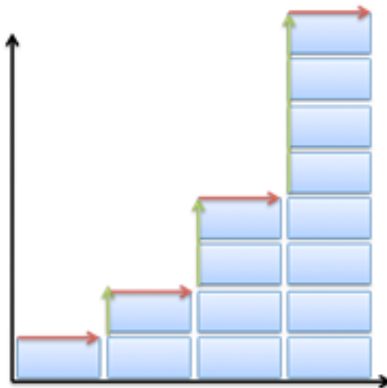
Qualitative Data	Quantitative Data
Overview: - Deals with descriptions. - Data can be observed but not measured. - Colors, textures, smells, tastes, appearance, beauty, etc. Qualitative → Quality	Overview: - Deals with numbers. - Data which can be measured. - Length, height, area, volume, weight, speed, time, temperature, humidity, sound levels, cost, members, ages, etc. Quantitative → Quantity

Scientific laws are short, sweet, and always true. They're often expressed in a single statement and generally rely on a concise mathematical equation. Laws are accepted as being universal and are the cornerstones of science. They must never be wrong (that is why there are many theories and few laws). If a law were ever to be shown false, any science built on that law would also be wrong.

Discrete

Values or observations can be counted (1,2,3,...) and are **distinct and separate**. Examples: the number of kittens in a litter; number of threads in a sheet, number of bricks in a wall.

Discrete Growth (2^n)

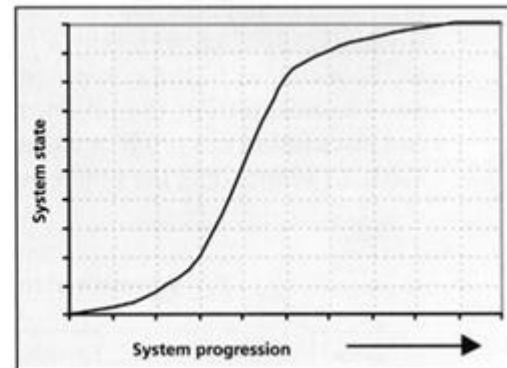
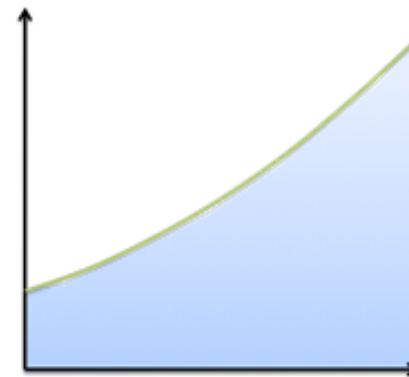


vs.

Continuous

You can **measure** continuous data. Values or observations **may take on any value** within a finite or infinite interval. Examples: height, time and temperature.

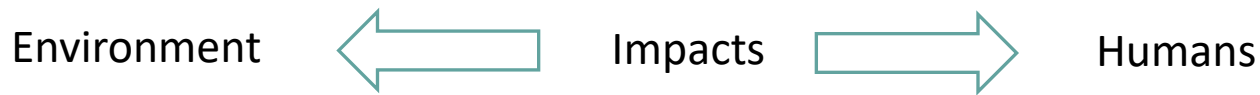
Continuous Growth (e^x)





- ❖ The nominal type, sometimes also called the **qualitative** type, differentiates between items or subjects based only on their names or (meta-) categories and other qualitative classifications they belong to. Examples include gender, nationality, ethnicity, language, genre, style, biological species, and form.
- ❖ The ordinal type allows for rank order (1st, 2nd, 3rd, etc.) by which data can be sorted, but still does not allow for relative *degree of difference* between them.

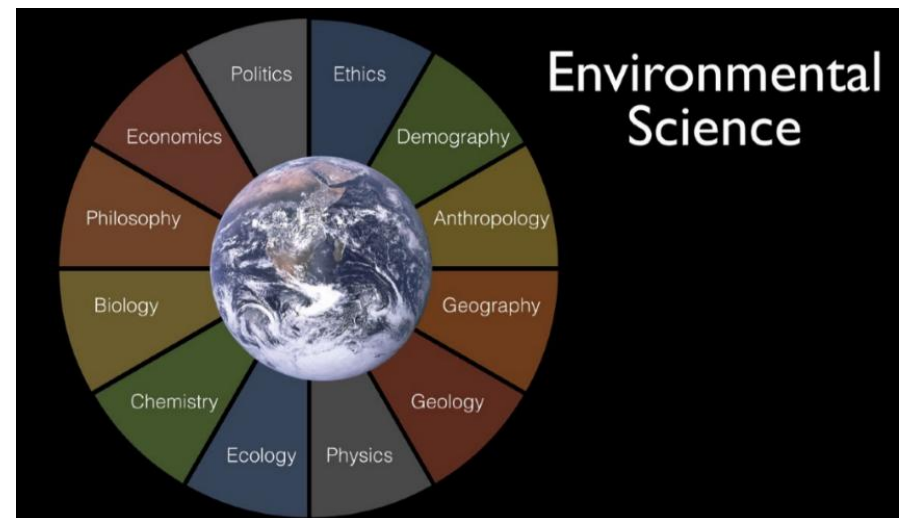
Environmental Science – How it Works



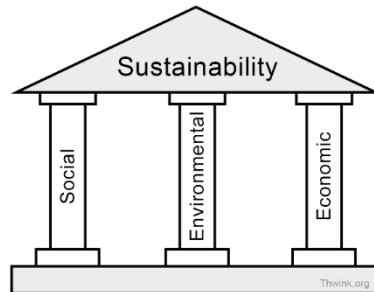
how nature works
how the environment affects us
how we affect the environment
how we deal with environmental problems
how we can live more sustainably without degrading our life-support system

Environmental science is a group of sciences that attempt to explain how life on the Earth is sustained, what leads to environmental problems, and how these problems can be solved.

Many sciences are important to environmental science. These include biology (especially ecology, that part of biology that deals with the relationships among living things and their environment), geology, hydrology, climatology, meteorology, oceanography, soil science etc. Also engineering sciences are required to deal with most of environmental problems.

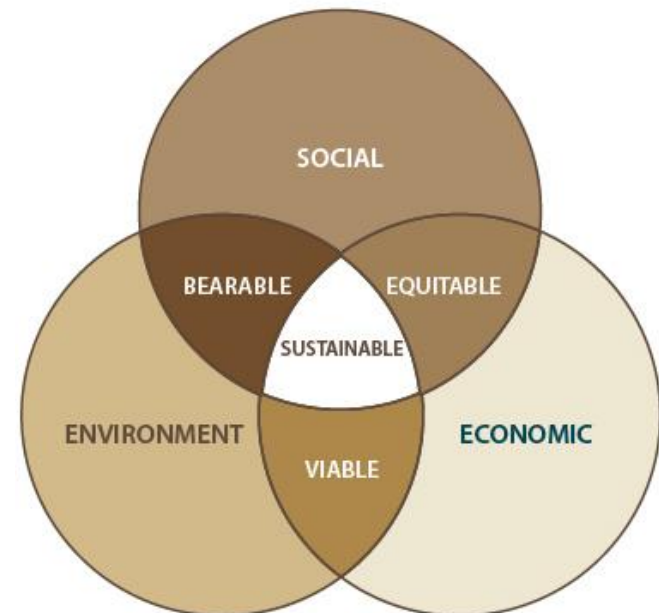
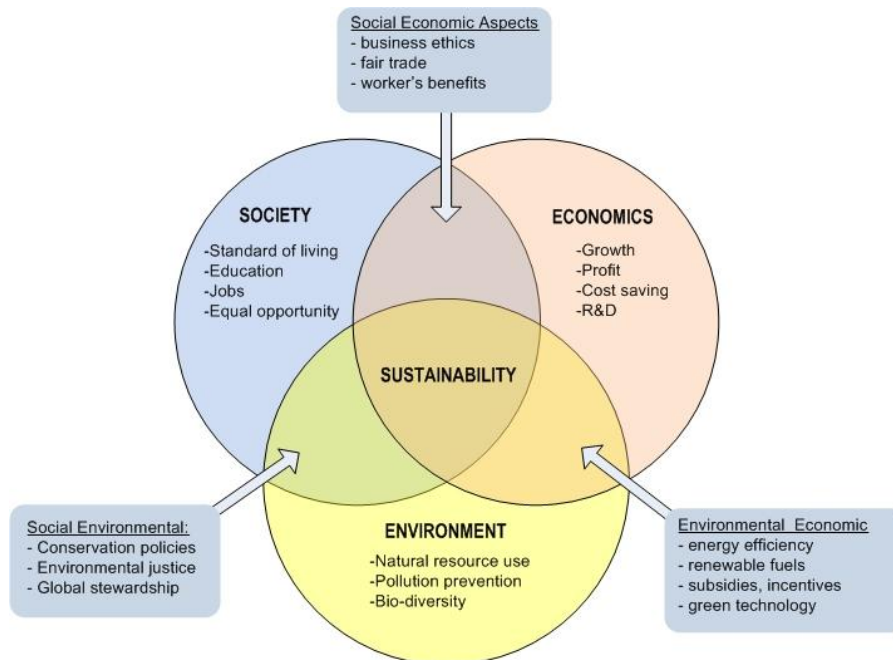


Three Spheres of Sustainability



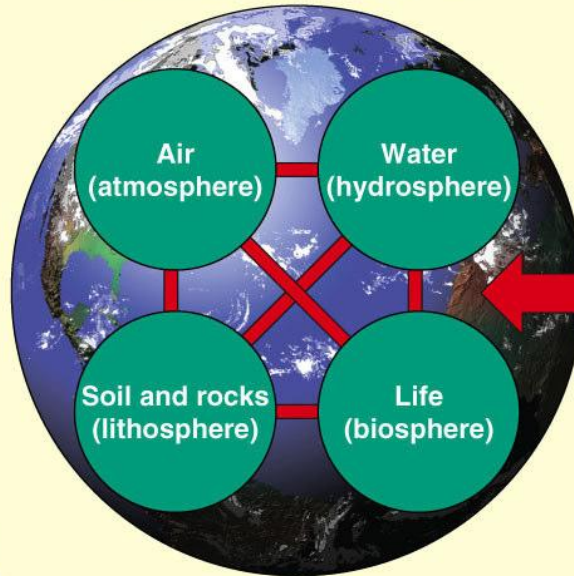
Environmental science includes also other fields that have to do with how we value the environment, from environmental philosophy to environmental economics.

Understanding environmental sciences requires, interdisciplinary studies in natural sciences, including geology, climatology, hydrology, ecology, **and their interaction** with social sciences such as economics, political science, sociology, anthropology, geography.

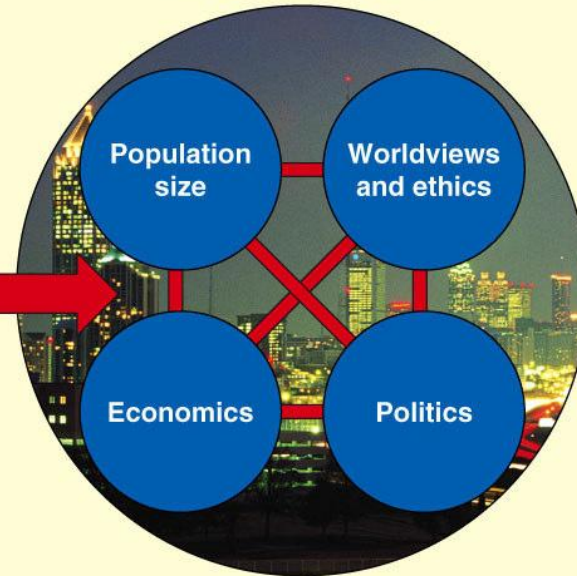


ENVIRONMENTAL SCIENCE

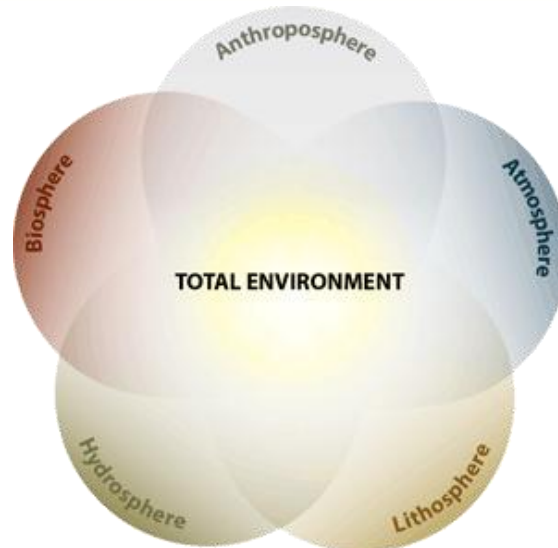
Earth's Life-Support System



Human Culturesphere



© 2007 Thomson Higher Education



Difference Between Ecology & Environmental Science?



ECOLOGY



Ecology is a sub-discipline of biology studying the relationships between organisms (biotic) and their environment (abiotic).

ENVIRONMENTAL SCIENCE

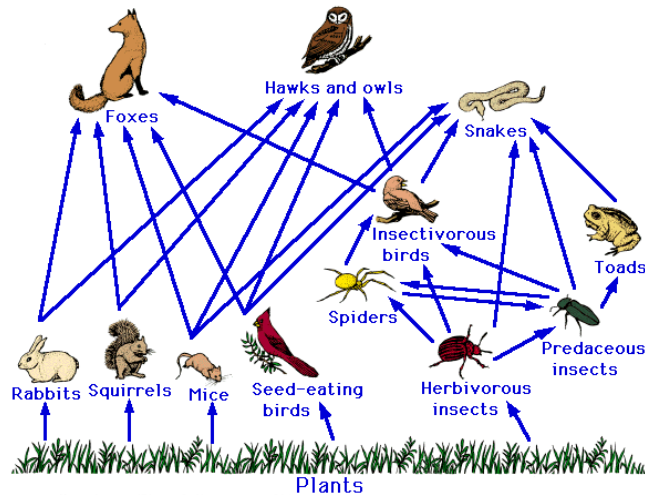


Environmental science is the study of the interactions among the physical, chemical and biological components of the environment; with a focus on pollution and degradation of the environment related to human activity; and the impact on biodiversity and sustainability from local and global development.

Ecology



Study of the interactions between organisms and their environment.

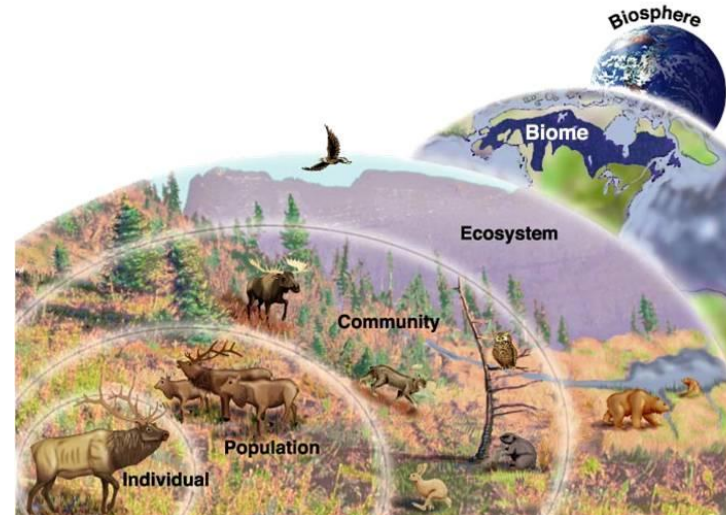


<http://marialombardic.blogspot.com/2010/10/food-chain-of-tiger.html>

Ecosystem



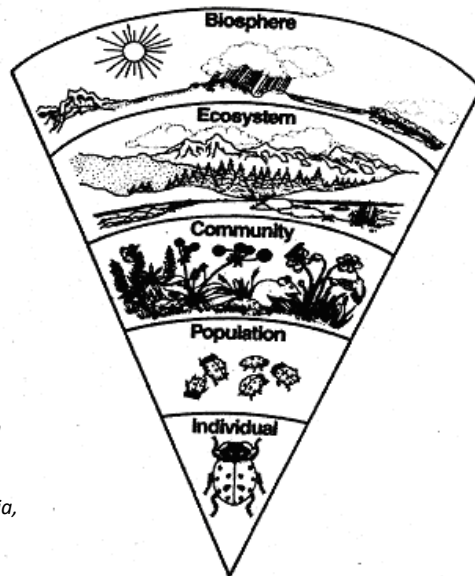
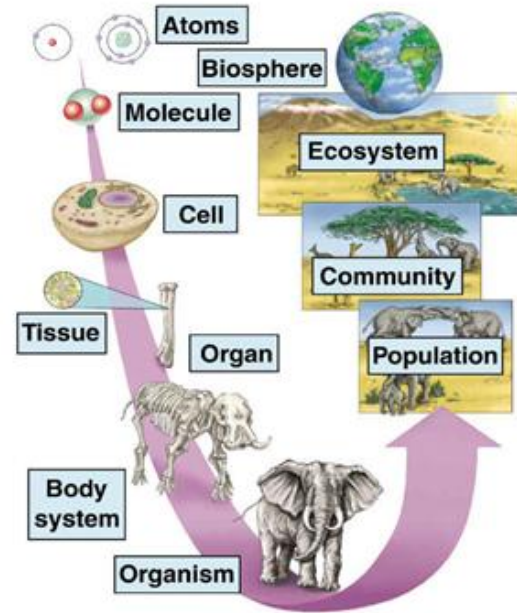
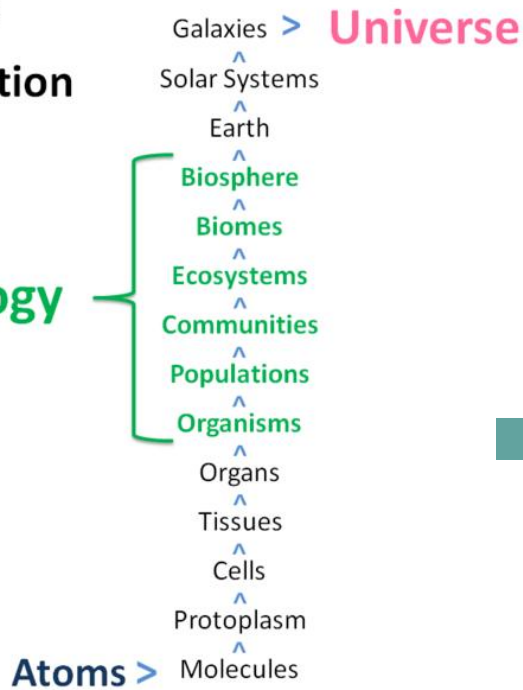
Includes all organisms living in an area and the physical environment which these organisms interact.



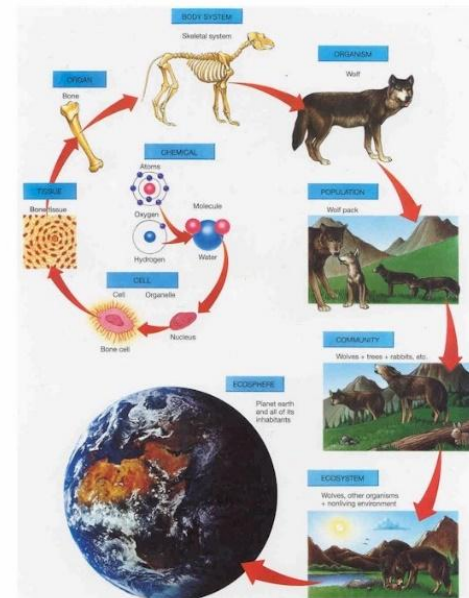
<http://www.goldiesroom.org/Note%20Packets/22%20Ecology/00%20Ecology--WHOLE.htm>

Levels of Organization

Ecology



Levels of organization in an ecosystem
(Source - Flint, ML and Gouveia, P. 2001)



Pollution

- ❖ An undesirable change in the physical, chemical, or biological characteristics of air, water, soil, or food that can adversely affect the health, survival, or activities of humans or other living organisms.
- ❖ When only **short-term economic gains** are made at the cost of **long-term ecological benefits** for the humanity.
- ❖ No natural phenomenon has led to greater ecological changes than have been made by mankind!

Pollution is the harmful alteration of our environment by our own actions.

Pollutants are either unwanted by-products or our activities or the obnoxious residues of things we have made, used, and thrown away.



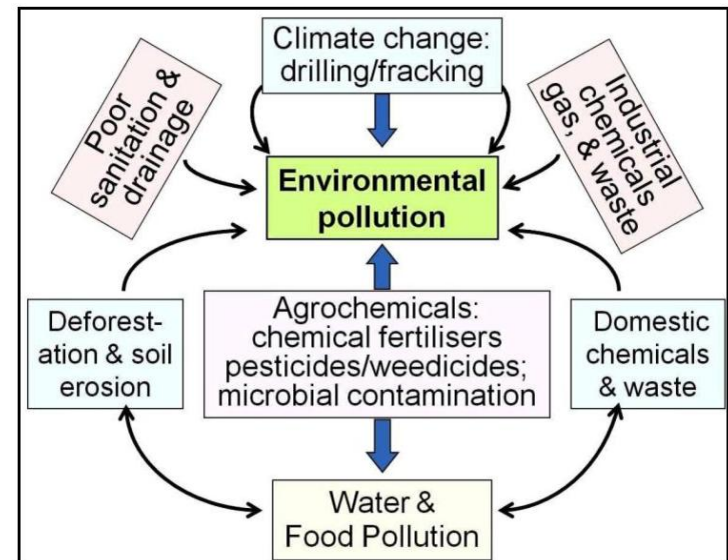
Pollution

❖ Pollutants can have three types of unwanted effects:

- Can disturb/degrade life-support systems
- Can damage health and property
- Can create nuisances such as noise and unpleasant smells, tastes and sights

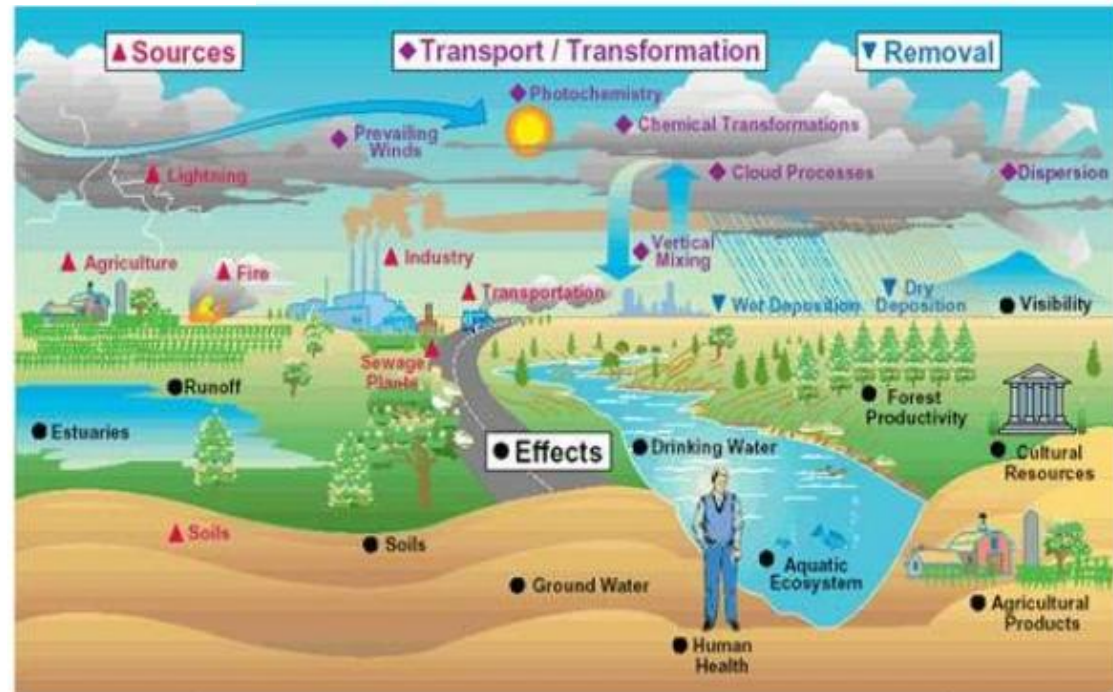
❖ Three factors determine how severe the harmful effects of a pollutant are:

- The concentration level (ppm, ppt, etc.)
- The nature of the chemical (How active and harmful it is in the nature)
- Half-life or the persistence of the pollutant (e.g. DDT)



Pollution

- Air pollution
 - Most air pollution is caused by the burning of fossil fuels.
- Water pollution
 - Organic sewage
 - Eutrophication
 - Infectious agents
 - Organic chemicals
 - Inorganic and miscellaneous chemicals
 - Sediments from land corrosion
 - Radioactive substances
 - Waste heat from power plants and industry
- Land pollution
 - Pesticides – chemicals used to kill insects defined as pests.
 - Herbicides – chemicals used to kill plant life, particularly weeds.
 - Chemical wastes
 - Radioactive fallout
 - Acid rain
 - Garbage



- Global warming
 - Since the late 1800's the average global surface temperature has increased about 0.75 degrees C.
 - Most warming has occurred since 1950.

Dealing with Pollution

Point Source

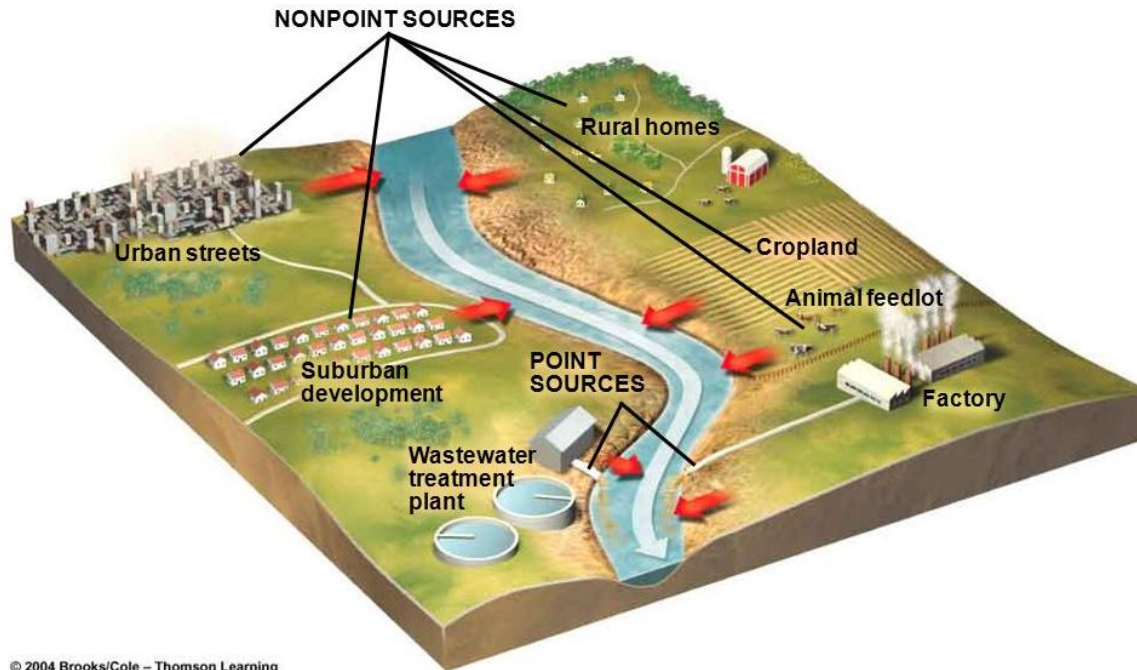


Single identifiable source that discharges pollutants into the environment. E.g. smokestack, exhaust pipes, industrial discharge.

Non-Point Source



Large or dispersed land areas such as crop fields, streets, and lawns that discharge pollutants into the environment over a large area. E.g. stormwater, septic tanks.

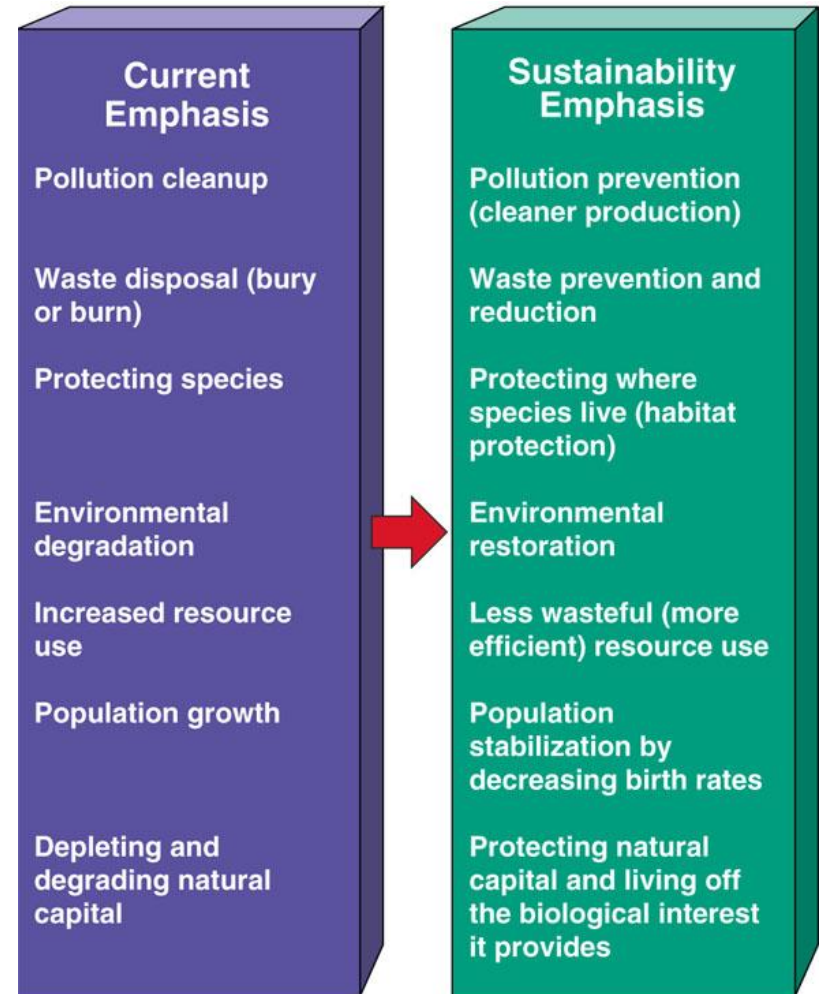


Solutions

- ❖ Prevention
 - input control
- ❖ Cleanup
 - output control
- ❖ Which strategy is more effective?
 - Why?
 - Where should we put more emphasis?

❖ Current Emphasis (Reactive)

❖ Sustainability Emphasis (Proactive)



© 2005 Brooks/Cole - Thomson

Environmental problems...

Dispersion of Toxic Substances

Water Pollution & Water Resources

Ozone Layer Depletion

Waste

Air Pollution

Climate Change

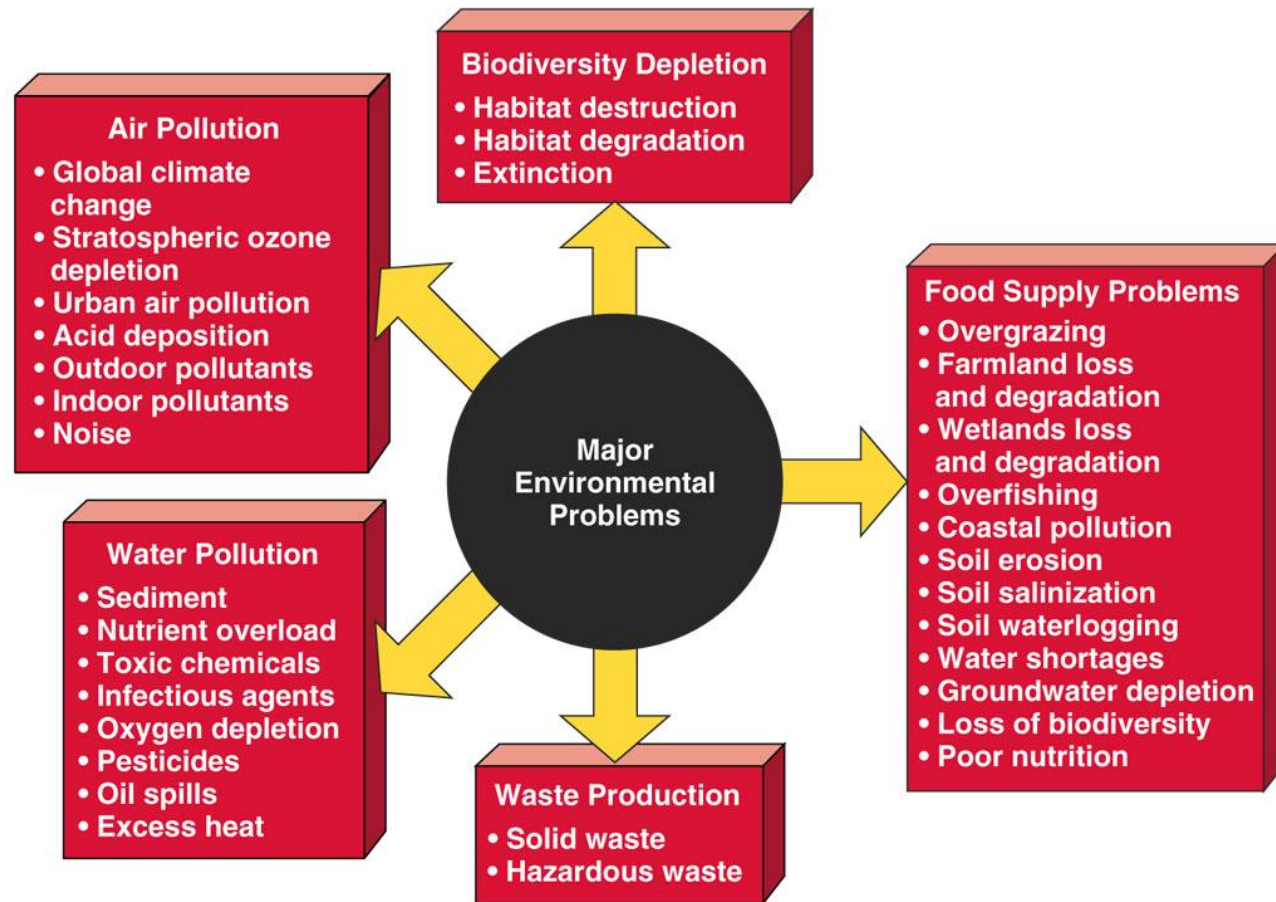
Urban Environmental Problems

Loss of Biodiversity

Marine Environment & Coastal Zones

Resource Depletion

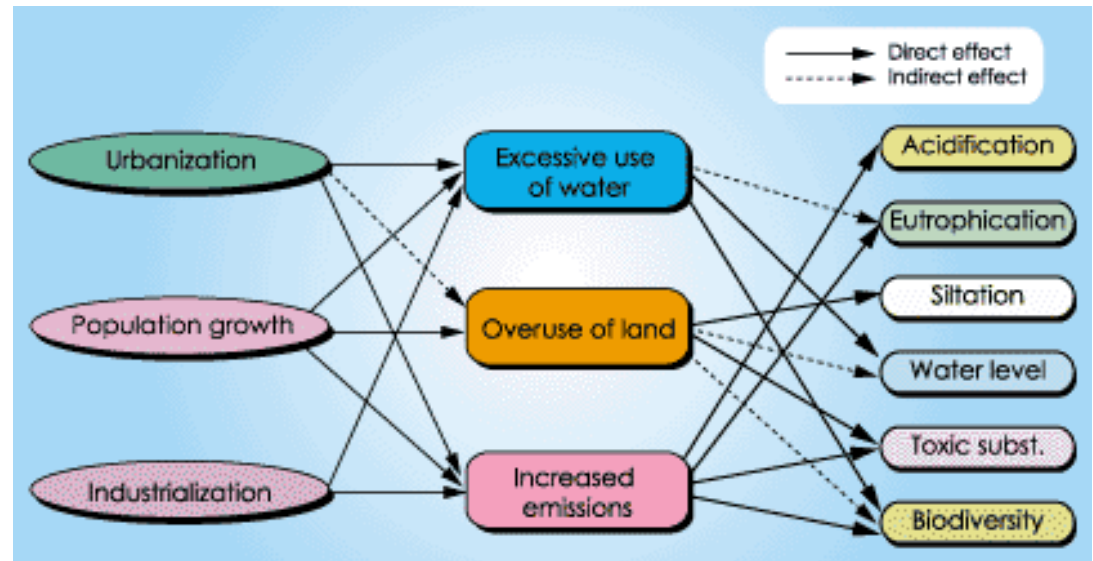
Environmental Resource Problems



What is Our Greatest Environmental Problem?

A lake example:

- ❖ Disease
- ❖ Overpopulation
- ❖ Water Shortages
- ❖ Climate Changes
- ❖ Biodiversity Loss
- ❖ Poverty
- ❖ Malnutrition



http://www.unep.or.jp/ietc/publications/short_series/lakereservoirs-3/2.asp

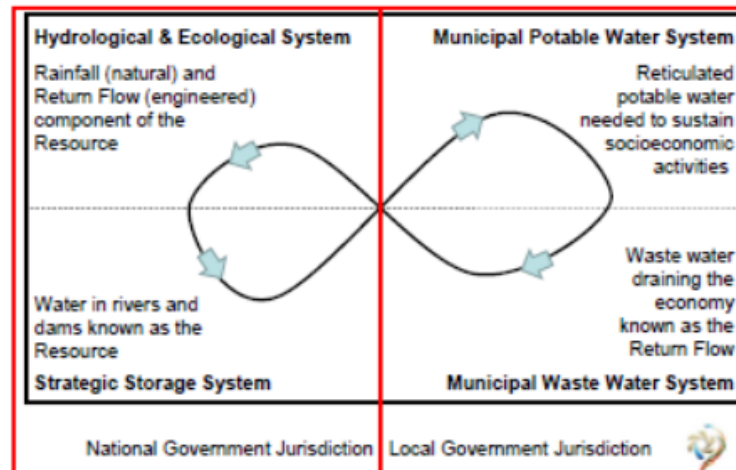
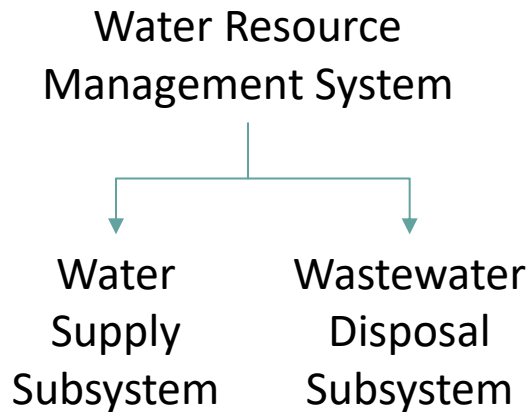
Urbanization, population growth and industrialization are amongst the basic factors which cause environmental problems in lakes and reservoirs.

What is Environmental Engineering?

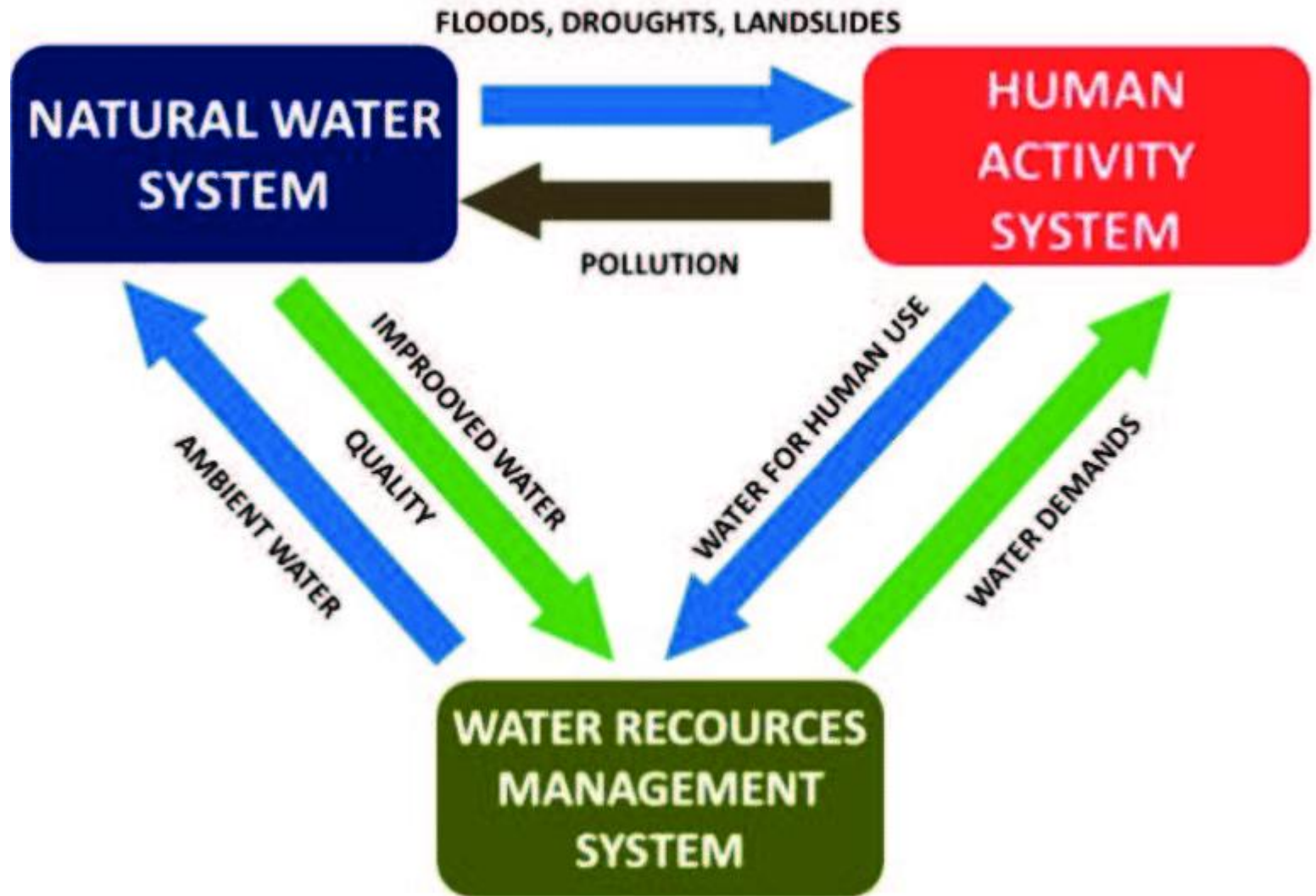
Environmental engineering is manifest by sound engineering thought and practice in the solution of problems of environmental sanitation, notably in the provision of safe, palatable, and ample public water supplies; the proper disposal of or recycle of wastewater and solid wastes; the adequate drainage of urban and rural areas for proper sanitation; and the control of water, soil, and atmospheric pollution, and the social and environmental impact of these solutions. Furthermore, it is concerned with engineering problems in the field of public health, such as control of arthropod-borne diseases, the elimination of industrial health hazards, and the provision of adequate sanitation in urban, rural, and recreational areas, and the effect of technological advances on the environment.



Environmental Systems Overview



- ❖ Pollution problems that are confined to one of these systems are called **single-medium problems** if the medium is either air, water, or soil.
- ❖ Many important environmental problems are not confined to one of these simple systems but cross the boundaries from one to another. Such problems are referred to as **multimedia pollution problems**.



A Little Environmental History: The Rise of Environmentalism

Before 1960, few people had ever heard the word *ecology*, and the word *environment* meant little as a political or social issue. Then came the publication of *Rachel Carson's* landmark book, *Silent Spring* (Boston: Houghton Mifflin, 1962). It is a classic book on problems associated with toxins in the environment. At about the same time, several major environmental events occurred, such as oil spills along the coasts of Massachusetts and southern California, and highly publicized threats of extinction of many species, including whales, elephants, and songbirds. The environment became a popular issue.

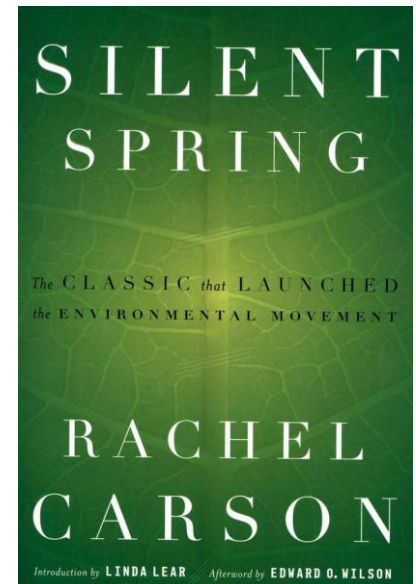
*'The more clearly
we can focus
our attention on
the wonders and
realities
of the universe
about us,
the less taste we
shall have
for destruction.'*
Rachel Carson,
1954



Rachel Carson (1907-1964)

Disturbed by the profligate use of synthetic chemical pesticides after World War II, Carson reluctantly changed her focus in order to warn the public about the long-term effects of misusing pesticides.

In *Silent Spring* (1962) she challenged the practices of agricultural scientists and the government and called for a change in the way humankind viewed the natural world.



Social movement for protecting earth's life support systems for us and other species

**Ecological, conservation, pollution
control organisations**

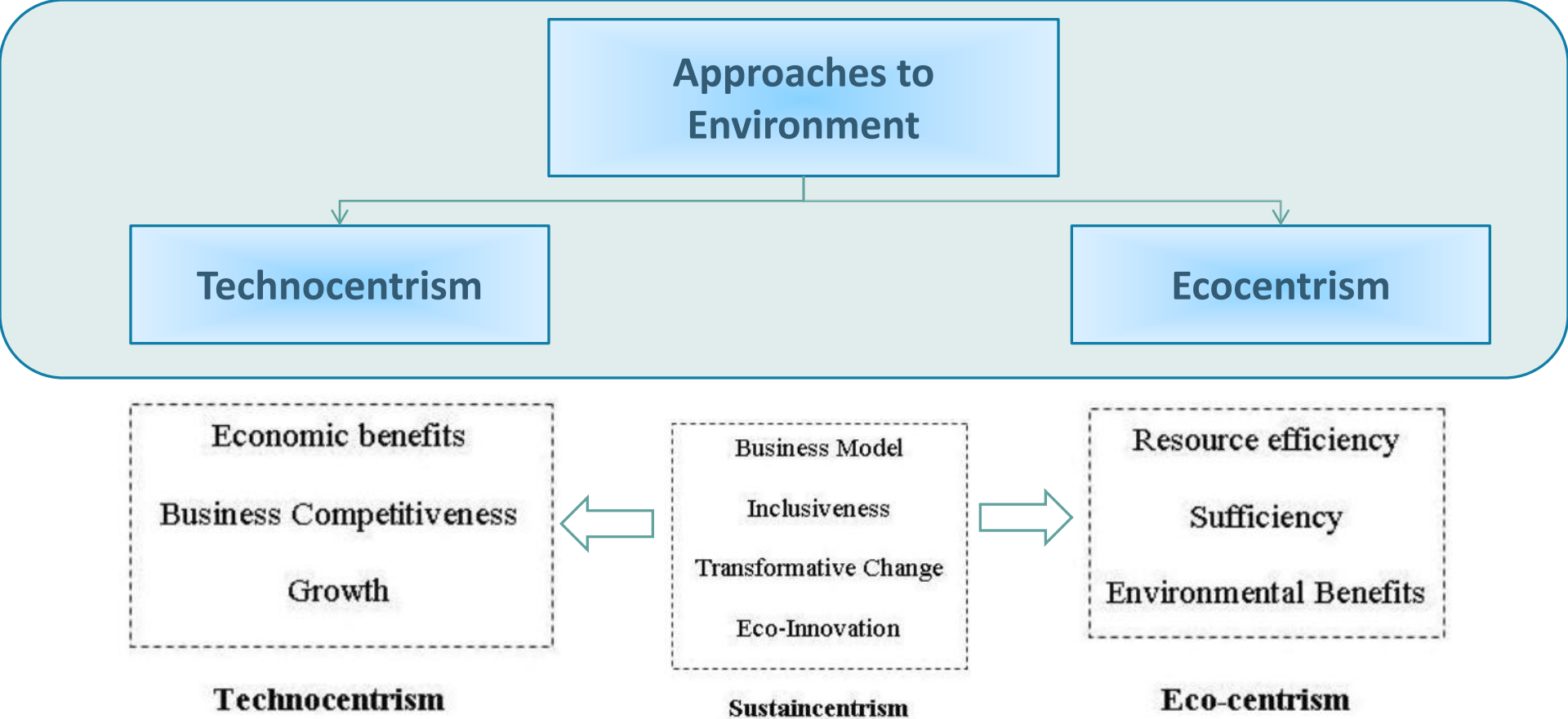
Environmentalism

Environmentalism

Social movement for protecting earth's life support systems for us and other species

Environmentalism, political and ethical movement that seeks to improve and protect the quality of the natural environment through changes to environmentally harmful human activities; through the adoption of forms of political, economic, and social organization that are thought to be necessary for, or at least conducive to, the benign treatment of the environment by humans; and through a reassessment of humanity's relationship with nature. In various ways, environmentalism claims that living things other than humans, and the natural environment as a whole, are deserving of consideration in reasoning about the morality of political, economic, and social policies.





Technocentrism: 'The nature of the physical world permits continued improvement in humankind's economic lot in the long run, indefinitely... Of course, there are always newly arising local problems but resilience in a well-functioning economic and social systems enables us to overcome such problems, and solutions usually leave us better off than if the problem had never arisen' (J. Simon and H. Kahn, 1984, *The Resourceful Earth*)

Ecocentrism: 'Ecology teaches us that humankind is not the centre of life on the planet. Ecology has taught us that the whole Earth is part of our 'body' and that we must learn to respect it as we respect ourselves' (Greenpeace)



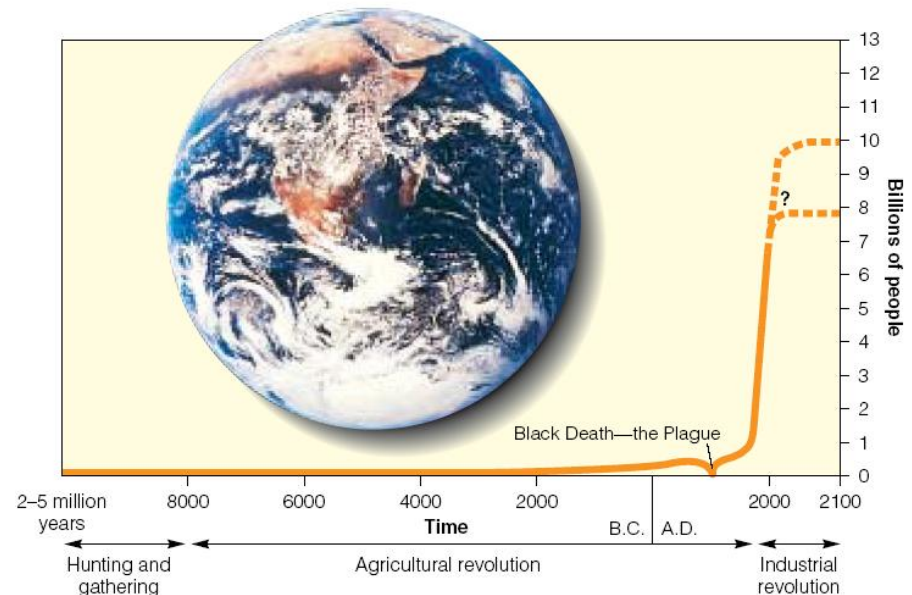
Sustainability means that a resource is used in such a way that it continues to be available.

Some would define it as follows:

We have the right to use the earth's resources and earth capital to meet our needs, but we have the obligation to pass on the earth's resources and services to future generations in as good or better shape than these conditions were passed on to us.

Population Growth & Earth's Carrying Capacity

The global population has risen dramatically during the last century. Underlying nearly all environmental problems is this rapid rise in human population.



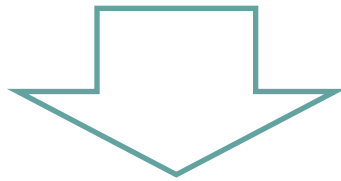
The Black Death, also known as the Great Plague, the Black Plague, or the Plague, was one of the most devastating pandemics in human history, resulting in the deaths of an estimated 75 to 200 million people in Eurasia and peaking in Europe from 1347 to 1351.

It is difficult to determine Earth's carrying capacity. Estimates of Earth's carrying capacity for people range from 2.5 to 40 billion, but about 15 billion is the upper limit with today's technology. The differences in capacity have to do with the quality of life projected for people—the poorer the quality of life, the more people can be packed onto the Earth.

Critical thinking issue:

What is more important?

The quality of your local environment or the quality of the global environment—the environment of the entire planet?



Placing a Value on the Environment

The value of the environment is based on five justifications

The utilitarian justification

Some aspect of the environment is valuable because it benefits individuals economically or is directly necessary to human survival. For example, conserving lions in Africa as part of tourism provides a livelihood for local people.

The ecological justification

An ecosystem is necessary for the survival of some species of interest to us, or that the system itself provides some benefit.

Aesthetic and recreational justification

Our appreciation of the beauty of nature and our desire to get out and enjoy it. For example, many people find wilderness scenery beautiful and would rather live in a world with wilderness than without it.

Moral justification

The belief that various aspects of the environment have a right to exist and that it is our moral obligation to help them, or at least allow them, to persist.

Cultural justification

Different cultures may have different values with respect to the environment. All cultures may value nature, but, depending on their religious beliefs, may value it in different degrees of intensity.

Analysis of environmental values is the focus of a new discipline, known as ***environmental ethics***.